

WattWise

An Intelligent Home Energy Management and Monitoring System

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Chapter 1 — Introduction

1.1 Background

Electricity consumption has become one of the primary concerns for modern households due to increasing energy demand and rising electricity costs. Although smart meters and prepaid electricity systems provide consumers with information about their total electricity usage, they do not reveal how individual appliances contribute to overall consumption. As a result, users often become aware of excessive energy usage only after receiving a high electricity bill or exhausting their prepaid balance.

In Egypt, the widespread adoption of prepaid smart electricity meters and tier-based electricity tariffs has increased the importance of monitoring and managing household energy consumption. However, existing solutions generally provide only overall consumption values without identifying which appliances are responsible for increased usage or offering actionable recommendations.

Recent advancements in the Internet of Things (IoT) enable the development of intelligent monitoring systems capable of collecting real-time electrical measurements, analyzing consumption patterns, and assisting users in making informed energy decisions. This project utilizes IoT technologies to build an intelligent, appliance-level energy management system tailored to the needs of Egyptian households.

1.2 Problem Statement

Many households face several challenges regarding electricity consumption:

- Lack of visibility into the electricity consumption of individual appliances.
- Difficulty identifying devices responsible for high energy usage.
- Limited awareness of abnormal electrical behavior that may indicate appliance degradation or wiring issues.
- Risk of exhausting prepaid electricity balance unexpectedly.
- Difficulty monitoring progress toward higher electricity tariff tiers.
- Absence of intelligent recommendations that help users optimize their electricity usage.

Consequently, users often react to increased electricity costs after they occur instead of preventing them through proactive monitoring and decision-making.

1.3 Proposed Solution

The proposed system is an IoT-based Home Energy Management and Monitoring System that continuously measures the electrical characteristics of household appliances using smart monitoring nodes installed at selected power outlets.

Each monitoring node measures voltage, current, power, and energy consumption of the connected appliance and transmits the collected data to a central management platform. The platform aggregates measurements from all monitored outlets, stores historical data, analyzes consumption patterns, and presents the information through an interactive dashboard.

Beyond monitoring, the system provides intelligent services including:

- Appliance-level energy monitoring.
- Whole-house energy analysis.

- Electricity cost estimation.
- Tariff tier tracking and prediction.
- Prepaid electricity balance management.
- Fault indication based on abnormal electrical behavior.
- Energy-saving recommendations.
- Real-time notifications and alerts.
- Optional remote control of selected appliances.

The objective is not only to measure energy consumption but also to help users understand, optimize, and manage their household electricity usage more effectively.

1.4 Objectives

General Objective

To develop an intelligent IoT-based system that monitors, analyzes, and optimizes household electricity consumption while assisting users in reducing energy costs and improving electrical awareness.

Specific Objectives

- Monitor voltage, current, power, and energy consumption of individual appliances.
- Calculate total household electricity consumption.
- Estimate electricity cost based on consumption.
- Support monitoring of Egypt's prepaid electricity system.
- Warn users before reaching higher electricity tariff tiers.
- Detect abnormal electrical behavior that may indicate appliance or wiring issues.
- Generate intelligent recommendations for improving energy efficiency.
- Provide real-time notifications for important events.
- Allow remote monitoring and optional control of connected appliances.
- Maintain historical records for analysis and visualization.

1.5 Scope

The first version of the system includes:

Included

- Smart monitoring nodes for household outlets.
- Real-time electrical measurements.
- Appliance-level dashboard.
- House-level dashboard.
- Historical data storage.
- Consumption analytics.
- Cost estimation.
- Tariff prediction.
- Notifications.
- Recommendation engine.

- Optional relay-based control.

Excluded

- Direct integration with the national electricity provider.
- Automatic reading of official electricity meters.
- Revenue-grade certified billing measurements.
- Industrial energy management.
- Three-phase electrical systems (prototype version).

1.6 Expected Benefits

The proposed system provides benefits for both users and researchers.

For Users

- Better understanding of electricity consumption.
- Lower electricity costs through informed decisions.
- Increased awareness of appliance behavior.
- Early indication of potential electrical problems.
- Better management of prepaid electricity balance.
- Reduced likelihood of entering higher tariff tiers unexpectedly.

For Researchers and Developers

- Demonstrates practical application of IoT technologies.
- Integrates embedded systems, cloud communication, databases, and analytics.
- Provides a scalable architecture for future smart-home applications.
- Serves as a foundation for advanced energy management research.

Chapter 2 — System Analysis and Requirements

2.1 System Overview

The proposed system is an IoT-based Home Energy Management and Monitoring System designed to provide real-time monitoring, analysis, and management of household electricity consumption.

The system consists of multiple smart monitoring nodes installed at selected electrical outlets throughout the house. Each node measures electrical parameters of the connected appliance and communicates with a central management platform via Wi-Fi.

The collected data are processed, stored, analyzed, and presented through a user-friendly dashboard. The system also generates intelligent recommendations, notifications, and warnings to help users optimize electricity consumption, reduce costs, and identify abnormal electrical behavior.

2.2 Stakeholders

Stakeholder	Role
Home Owner	Monitors and manages household electricity usage.

Stakeholder	Role
Family Members	Use connected appliances and receive notifications if authorized.
System Administrator	Configures devices, monitors system status, and manages users.
IoT Monitoring Nodes	Collect electrical measurements and communicate with the server.
Backend Server	Processes data, performs analytics, and manages automation.

2.3 Functional Requirements

Monitoring — the system shall:

- Measure outlet voltage.
- Measure outlet current.
- Calculate instantaneous power.
- Calculate accumulated energy consumption.
- Monitor appliance operating status.
- Record timestamps for all measurements.

Data Management — the system shall:

- Store historical measurements.
- Organize data by appliance.
- Organize data by room.
- Organize data by household.
- Retrieve historical records.
- Support filtering by time period.

Dashboard — the dashboard shall display:

Appliance Dashboard:

- Outlet number / name
- Voltage
- Current
- Power
- Energy consumed
- Daily cost
- Monthly cost
- Historical charts
- Device status

House Dashboard:

- Total power
- Total energy
- Current tariff tier
- Remaining consumption before next tariff

- Highest consuming appliance
- Peak consumption period
- System notifications

Notifications — the system shall notify users when:

- Consumption exceeds a defined threshold.
- An appliance behaves abnormally.
- The prepaid balance becomes low.
- The household approaches the next tariff tier.
- Communication with a monitoring node is lost.
- A relay changes state.

Recommendations — the system shall generate recommendations such as:

- Reduce usage of the highest consuming appliance.
- Delay operating heavy loads to avoid reaching the next tariff tier.
- Disconnect appliances drawing excessive standby power.
- Investigate appliances exhibiting unusual electrical behavior.
- Reduce overall consumption when prepaid credit is low.

Control — authorized users shall be able to:

- Turn supported appliances ON.
- Turn supported appliances OFF.
- Schedule appliance operation.
- Enable or disable automation rules.

2.4 Constraints

- Stable Wi-Fi connection is required.
- Monitoring accuracy depends on the selected sensing hardware.
- Prototype supports only monitored outlets.
- Official electricity meter data is not accessed directly.
- Cost calculations are estimates and depend on tariff configuration.

Chapter 3 — System Architecture

3.1 Architectural Overview

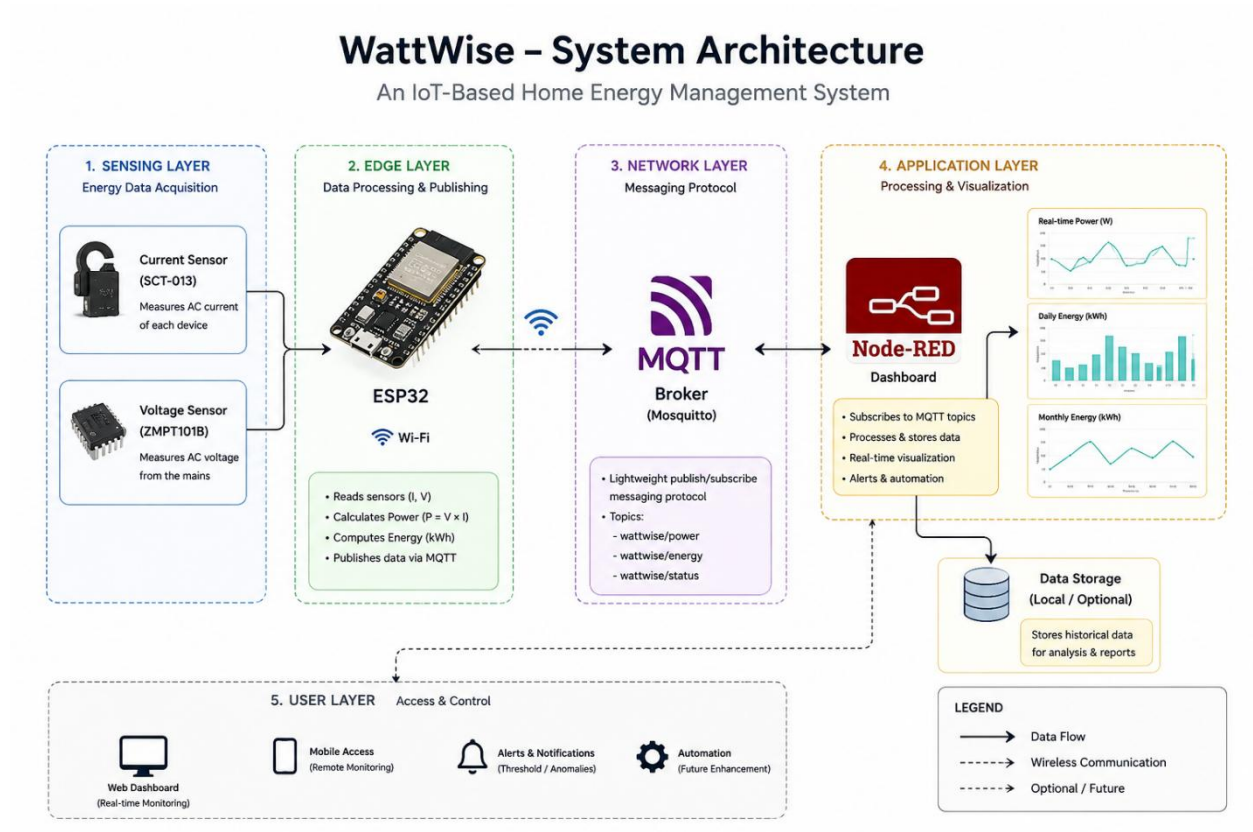
The proposed system follows a layered IoT architecture that separates data acquisition, communication, processing, storage, analytics, and user interaction. This modular design improves scalability, maintainability, and allows future expansion without redesigning the entire system.

The system consists of three intelligence levels:

- Outlet Intelligence — individual smart monitoring nodes collect electrical measurements and monitor connected appliances.

- Home Intelligence — the backend aggregates data from all monitoring nodes to analyze household energy consumption.
- User Intelligence — the dashboard converts collected information into visual insights, recommendations, and actionable notifications.

3.2 Overall Architecture



3.3 Layered Architecture

Layer 1 — Physical Layer

Responsible for interacting with the real world. Components include:

- Household appliances
- Electrical outlets
- Household wiring

Examples:

- Air conditioner
- Refrigerator
- Washing machine
- Television
- Water heater
- Lighting

Layer 2 — Sensing Layer

Responsible for measuring electrical parameters. Each monitoring node measures:

- Voltage
- Current
- Power
- Energy
- Relay status

Future versions may also measure:

- Frequency
- Power factor
- Apparent power
- Reactive power

Layer 3 — Edge Computing Layer

The microcontroller performs:

- Sensor acquisition
- Basic filtering
- Energy calculations
- Fault checking
- MQTT communication
- Relay control

Future hardware revisions may replace the ESP32 with a lower-power microcontroller while preserving the same software architecture.

Layer 4 — Communication Layer

Communication technologies include:

- Wi-Fi
- MQTT
- HTTP/HTTPS (API communication)

Responsibilities:

- Reliable message delivery
- Device identification
- Command distribution
- Remote monitoring

Layer 5 — Analytics and Intelligence Layer

Cost Analysis:

- Daily cost
- Monthly cost
- Prepaid balance estimation

Tariff Analysis:

- Current tariff tier

- Remaining consumption before next tier
- Predicted tariff transition

Fault Indication — the system identifies unusual electrical behavior such as:

- Abnormal current increase
- Unexpected voltage drop
- Excessive standby consumption
- Long operating periods
- Unusual power fluctuations

Recommendation Engine — examples:

- Reduce usage of Appliance X.
- Disconnect standby devices.
- Delay heavy loads.
- Recharge prepaid electricity soon.
- Inspect Appliance Y due to abnormal behavior.

Chapter 4 — Hardware Design

4.1 Hardware Overview

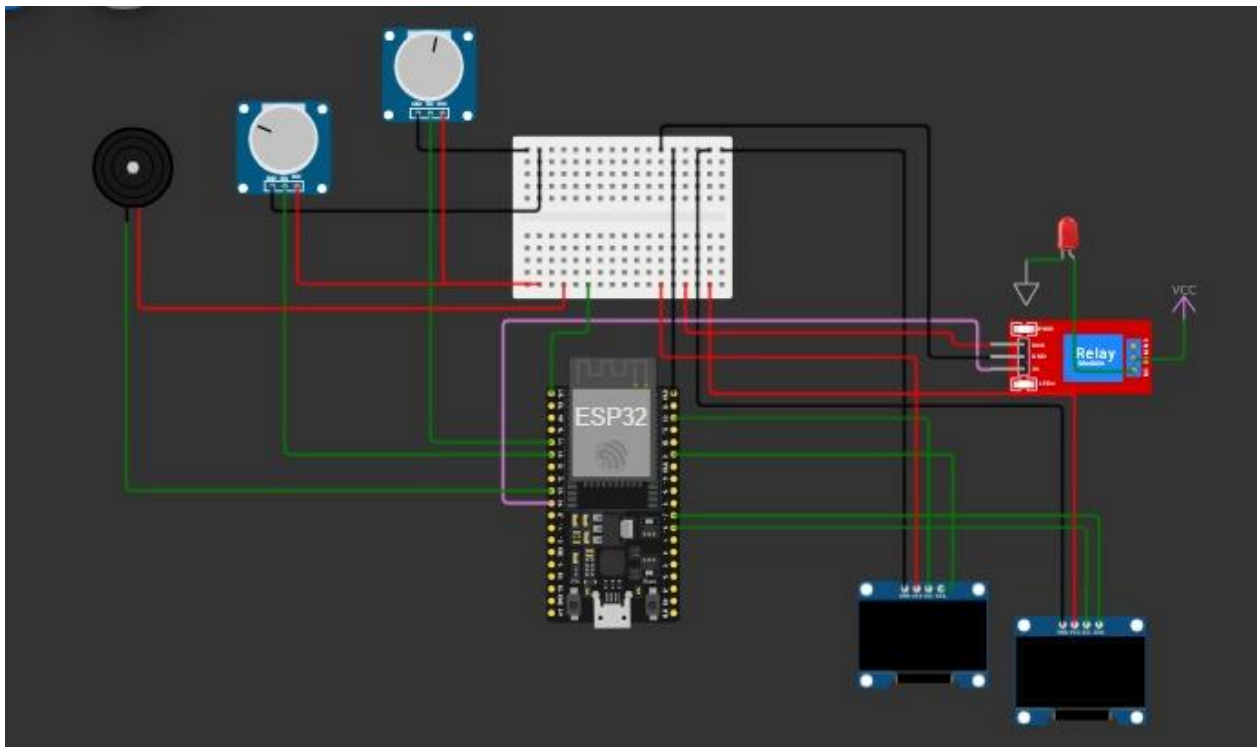
The proposed hardware consists of multiple Smart Monitoring Nodes installed at selected electrical outlets throughout the house.

Each node is responsible for:

- Measuring electrical parameters.
- Calculating energy consumption.
- Communicating with the central server.
- Optionally controlling the connected appliance.

The hardware is designed to be modular, allowing additional monitoring nodes to be added without modifying the overall system architecture.

4.2 Hardware Architecture



4.3 Main Hardware Components

Component	Function
Microcontroller	Reads sensors, processes data, communicates with backend
Voltage Sensor	Measures outlet voltage
Current Sensor	Measures appliance current
Relay Module	Controls appliance power (optional)
Power Supply	Converts AC mains to regulated DC voltage
Wi-Fi Module	Connects node to the local network
Status LEDs	Indicates power, Wi-Fi, and relay status (prototype only)

4.4 Microcontroller Selection

The prototype uses an ESP32 because it offers:

- Built-in Wi-Fi
- Sufficient processing power
- Multiple ADC/GPIO interfaces
- Large software ecosystem
- Rapid development

However, the final commercial version is not limited to ESP32. Future versions may use a lower-cost, lower-power microcontroller with only the required peripherals to reduce manufacturing cost and energy consumption.

4.5 Electrical Measurements

Each Smart Monitoring Node measures:

Parameter	Purpose
Voltage (V)	Detect supply quality and voltage fluctuations
Current (A)	Measure appliance load
Power (W)	Determine instantaneous consumption
Energy (kWh)	Calculate accumulated energy usage

These measurements form the basis for analytics, cost estimation, and fault indication.

4.6 Relay Module

Certain appliances may be remotely controlled using a relay. Possible operations include:

- Manual ON/OFF
- Scheduled operation
- Automation rules

4.7 Communication

Each monitoring node communicates through Wi-Fi using MQTT.

Data transmitted include:

- Device identifier
- Timestamp
- Voltage
- Current
- Power
- Energy
- Relay status

Commands received include:

- Relay ON
- Relay OFF
- Configuration updates

4.8 Prototype vs. Commercial Product

Prototype	Commercial Version
ESP32 development board	Custom embedded controller
Breadboard / prototype PCB	Dedicated PCB
General-purpose sensors	Energy metering ICs
External relay module	Integrated relay circuit
Manual assembly	Mass-manufactured hardware

Prototype	Commercial Version
Software-focused optimization	Cost and power optimization

Chapter 5 — Software Architecture

5.1 Software Overview

The software system is responsible for collecting electrical measurements from all Smart Monitoring Nodes (SMNs), processing the received data, storing historical records, analyzing consumption patterns, generating recommendations, and presenting information through an interactive dashboard.

The software follows a modular architecture, where each component has a clearly defined responsibility. This approach improves maintainability, scalability, and future extensibility.

5.2 Software Components

Module	Responsibility
Firmware	Reads sensors and communicates with the backend
MQTT Service	Transfers data between nodes and server
Backend API	Handles requests from the dashboard
Analytics Engine	Processes energy data
Recommendation Engine	Generates user recommendations
Notification Service	Sends alerts and warnings
Automation Service	Executes automation rules
Database	Stores all system data
Dashboard	Displays information to users

Chapter 6 — Communication Design

6.1 Communication Overview

The Smart Energy Advisor system adopts a publish/subscribe communication model using the MQTT protocol.

Each Smart Monitoring Node (SMN) periodically publishes electrical measurements to the MQTT broker. The backend subscribes to these topics, processes the incoming data, stores them in the database, and generates analytics, recommendations, and notifications.

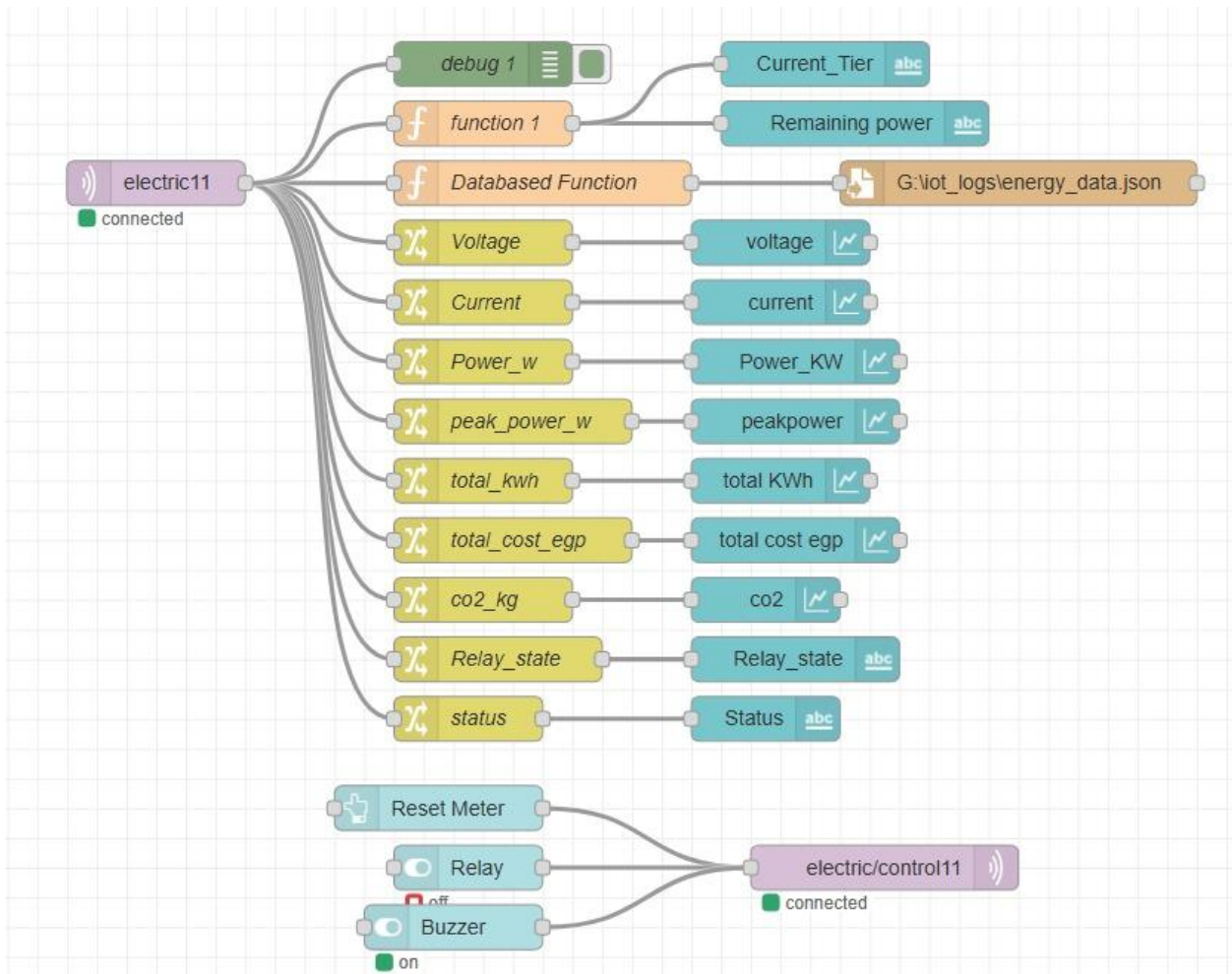
Similarly, the backend publishes control commands to the appropriate monitoring node, allowing remote control and configuration.

6.2 Why MQTT?

MQTT was selected because it is well suited for IoT applications due to:

- Lightweight communication with low bandwidth usage.
- Publish/Subscribe architecture.
- Low power consumption.
- Reliable message delivery through QoS.
- Scalability for hundreds or thousands of devices.
- Simple integration with cloud platforms.

6.3 Communication Architecture



6.4 MQTT Topic Structure

Instead of random topic names, the system uses a hierarchical naming convention:

```
energyadvisor/  
house/  
  {houseId}/  
  node/  
    {nodeId}/
```

This makes it easier to support multiple houses and users.

Measurement Topic

energyadvisor/house/{houseId}/node/{nodeId}/measurements

Published by: Smart Monitoring Node | Subscribed by: Backend

Status Topic

energyadvisor/house/{houseId}/node/{nodeId}/status

Example values: Online, Offline, Booting, Error

Command Topic

energyadvisor/house/{houseId}/node/{nodeId}/commands

Published by: Backend | Subscribed by: Smart Monitoring Node

Configuration Topic

energyadvisor/house/{houseId}/node/{nodeId}/config

Used to update: sampling interval, device name, thresholds.

Log Topic

energyadvisor/house/{houseId}/node/{nodeId}/logs

Useful during development and debugging.

6.5 Device Registration

When a Smart Monitoring Node starts for the first time:

```

Power ON
|
Connect Wi-Fi
|
Connect MQTT
|
Publish Registration
|
Receive Configuration
|
Begin Monitoring

```

The backend stores the node and associates it with the selected outlet.

6.6 Quality of Service (QoS)

Message Type	QoS
Measurements	QoS 0
Commands	QoS 1
Configuration	QoS 1
Registration	QoS 1
Critical Alerts	QoS 2 (optional)

Reasoning:

- Measurements are continuous, so occasionally losing one is acceptable.
- Commands and configuration updates should be delivered reliably.
- Critical alerts may justify the additional overhead of QoS 2.

6.7 Communication Sequence

Measurement Flow

Smart Monitoring Node → Read Sensors → Calculate Power
 → Publish MQTT Message → Broker → Backend
 → Database → Analytics → Dashboard

Control Flow

User → Dashboard → Backend → MQTT Broker → Node → Relay → Appliance

Chapter 7 — Future Work

The current implementation demonstrates the feasibility of an intelligent IoT-based home energy management system. However, several enhancements can be incorporated in future versions to increase functionality, scalability, and commercial value.

7.1 Smart Meter Integration

- Direct communication with official smart electricity meters.
- Automatic retrieval of prepaid balance and remaining credit.
- Real-time synchronization with actual electricity consumption.

7.2 Artificial Intelligence

Develop AI-based analytics capable of:

- Detecting abnormal appliance behavior automatically.

- Predicting future energy consumption.
- Forecasting monthly electricity costs.
- Identifying appliances with declining efficiency.
- Learning household energy usage patterns to generate personalized recommendations.

7.3 Mobile Application

Develop Android and iOS applications that provide:

- Real-time monitoring.
- Push notifications.
- Appliance control.
- Voice commands.
- Energy reports.

7.4 Renewable Energy Integration

Extend the platform to monitor and optimize:

- Solar photovoltaic systems.
- Battery energy storage systems.
- Home generators.
- Electric vehicle chargers.

This would transform the system into a comprehensive Home Energy Management System (HEMS).

7.5 Cloud-Based Deployment

Implement a cloud architecture to support:

- Multiple households.
- Remote access from anywhere.
- Centralized data storage.
- Fleet management for service providers.
- Automatic firmware updates.

7.6 Advanced Automation

- Weather-aware recommendations.
- Automatic load scheduling.
- Smart demand-response strategies.

7.7 Enhanced Hardware

- Designing a custom PCB.
- Reducing power consumption.
- Using dedicated energy metering ICs for higher accuracy.
- Integrating temperature monitoring for safety.
- Reducing manufacturing cost for commercial production.

7.8 Commercial Expansion

The platform can be extended beyond residential homes to:

- Offices.
- Schools.
- Hospitals.
- Hotels.
- Industrial facilities.
- Commercial buildings.

The modular architecture enables the system to scale from a single home to large multi-building deployments with minimal architectural changes.

Chapter 8 — Conclusion

This project presented the design and implementation of the Smart Energy Advisor, an IoT-based Home Energy Management System that enables real-time monitoring, analysis, and optimization of household electricity consumption.

The system combines Smart Monitoring Nodes, MQTT communication, a backend server, a database, and an interactive dashboard to provide users with detailed appliance-level insights, estimated energy costs, prepaid electricity tariff awareness, notifications, and intelligent recommendations.

Unlike conventional energy monitoring solutions, the proposed system emphasizes actionable decision support rather than simple data visualization. By identifying high-energy appliances, warning users before reaching higher electricity tariff tiers, and detecting abnormal electrical behavior, the system helps users reduce costs, improve energy efficiency, and increase awareness of their electricity usage.

The modular hardware and software architecture also allows future expansion through artificial intelligence, renewable energy integration, cloud connectivity, and advanced automation. As a result, the Smart Energy Advisor serves not only as a successful IoT prototype but also as a scalable foundation for future smart home energy management solutions.